

Spatially-unconstrained 3D Display: Dynamic Projection Mapping on Manipulable Rotating Objects

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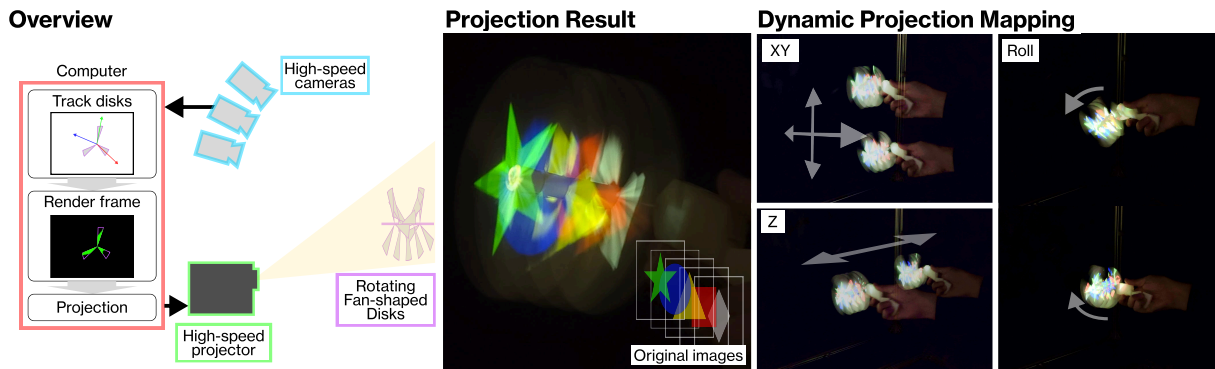


Figure 1: Left: System overview. The computer renders frames fit to the shape of the rotating object tracked by the high-speed camera and projects them using the high-speed projector. Center: Projection results on the rotating objects. Right: Through dynamic projection mapping, users can freely manipulate the rotating object to view the volumetric image from different angles.

ABSTRACT

This paper proposes a method for displaying 3D layered images through dynamic projection mapping onto a rotating layered fan. By utilizing high-speed tracking and projection techniques, we project images in synchronization with multi-layered rotating disks, creating the perception of layered volumetric images. Users can freely manipulate the rotating object's position and posture to observe the volumetric images from various positions and angles. This method enables everyday rotating objects, such as portable fans or DJ turntables, to be displayed and control volumetric graphics. In this paper, we detail the proposed method and the prototype implementation. Through demonstrations of applications that manipulate volumetric images, we discuss the usefulness and limitations of our proposed approach.

Index Terms: Dynamic projection mapping, tracking, high-speed projection, volumetric display.

1 INTRODUCTION

Conventional planar displays have limitations in expressing 3D computer graphics. To overcome these limitations, various systems capable of rendering more realistic and immersive graphics have been researched and developed. These systems are utilized to increase spatial perception and immersion in fields such as 3D modeling, room preview and entertainment. Various approaches have been proposed, including optical methods utilizing parallax and HMD-based methods. Among these, projecting specific images or structured light onto the surface of rapidly moving objects enables volumetric images to multiple viewers simultaneously without requiring special equipment [4, 5, 1, 3]. Furthermore, by optimizing the shape and material of rotating objects, more realistic images can

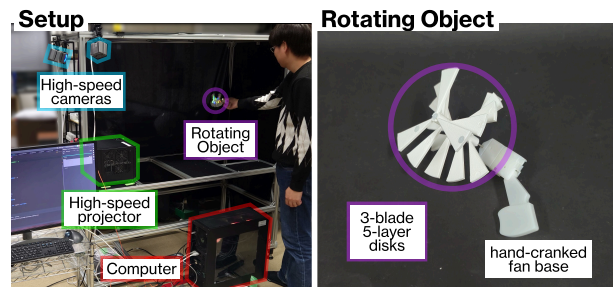


Figure 2: System setup. Left: System setup consisting of a high-speed camera, high-speed projector, control computer, and the rotating object as a projection target. Right: Close-up view of the rotating object, which comprises five fan-shaped disks with three blades each, mounted with phase offsets on a hand-crank fan base.

be projected [2, 6]. However, these methods require strict control of object movements and synchronization between the projection system and moving objects. Additionally, it is challenging to freely manipulate these systems for observation of volumetric graphics or system operation.

In this paper, we propose a method for observing 3D volumetric images by utilizing high-speed tracking to project images onto rotating stacked layered fans. By performing dynamic projection mapping onto this structure, different images are projected onto each rotating layered fan and viewers can see layered volumetric images in the rotating object. By utilizing vision tracking, the system can operate with inexpensive motors or even hand-cranked fans whose rotation speed is not strictly controlled. This enables users to freely move the rotating object. Consequently, our approach facilitates the use of everyday rotating objects as projection targets and enables user interaction with projected images.

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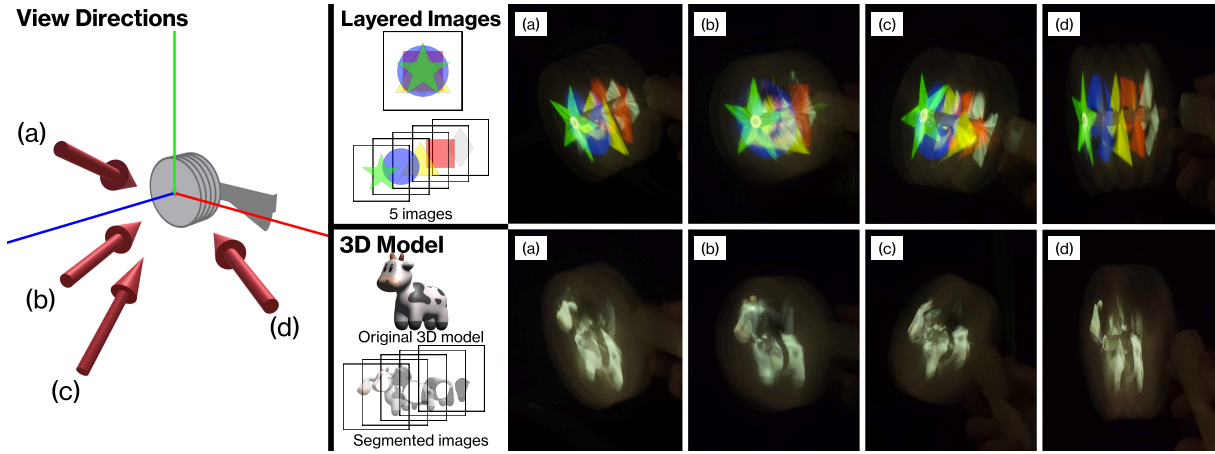


Figure 3: Projection results. Left: Result images are captured from four different angles labeled (a), (b), (c), and (d). Top right: Results of projecting five images with different colors and shapes onto each disk. Bottom right: Results of projecting five segmented images from a 3D model onto each disk.

2 PROTOTYPE SYSTEM

2.1 Method

The proposed method utilizes high-speed tracking and projection technologies to project images onto a layered fan-shaped rotating object. The processing pipeline is illustrated in Figure 3 (left). The system tracks the rotating object and processes target images by cropping them to match the object's shape. The system then projects the processed images onto the actual rotating object using a projector. By performing this sequence of processes at high speed, the target image appears on the rotating object through the persistence of vision effect.

2.2 Implementation

In this section, we describe the prototype implementation of the proposed method. The system consists of four main components: a rotating layered fan, a high-speed camera, a high-speed projector, and a computer for processing applications (Figure 2)

The rotating object consists of a three-blade, five-layer fan structure, where fan-shaped blades are stacked constant intervals with shifts of $\frac{1}{3 \times 5} \pi$ radians. We fabricated two variants of layered fans, a standard version and a thin version, for different experimental conditions. This configuration ensures that the blades do not visually overlap when viewed from the front. The layered fans are attached to the base of the commercially available hand-crank fan, and the user can grip it and turn the layered fans.

Our system employs three PrimeX 13W¹ high-speed cameras along with Motive² software to track the rotating layered fan with retro-reflective stickers at up to 240 FPS. We utilize openFrameworks³ for transform calculation and high-speed projection image rendering. Using this framework, we calculate the position and rotation of the layered fan from tracking data and render projection frames in real-time according to the rotation angle. The system incorporates DynaFlash⁴ for high-speed color projection. The projection image data is transmitted using a dedicated API.

To address the inherent system latency from tracking to projection, we implement predictive processing that accounts for rotational velocity around the axis.

¹<https://www.optitrack.jp/products/camera/primex13w.html>

²<https://www.optitrack.jp/products/software/>

³<https://openframeworks.cc/ja/>

⁴<https://www.inrevium.com/product/tb-6v-dynaflash/>

2.3 Result

Figure 3 shows the projection results. These images were captured using slow shutter speeds to reproduce the visual perception of afterimages. In the layered image demonstration, the system projected five shapes with different colors and shapes onto each layer. The resulting images show that the system successfully displayed layered visuals. In the 3D model demonstration, the system projected the 3D model's segmented images along its depth onto each layer. As a result, the system successfully expressed 3D models. These 3D images remain stable while users interact with the rotating object, as shown in Figure 1 (right).

3 DISCUSSION AND LIMITATIONS

Through our prototype implementation, we have demonstrated that the proposed method can project a volumetric image onto the rotating object while enabling free user interaction.

The proposed method has two significant challenges. The first challenge is handling projection latency, which causes misalignment between the rotating layered fans and the projected image. To address this problem, we can use higher-speed cameras to reduce total latency or implement more accurate rotation speed prediction algorithms. The second challenge concerns projection occlusion between projection blades. Depending on the projection direction, the upper blade casts shadows on lower blades, degrading the quality of the volumetric image. One potential solution is to use multiple projectors to compensate for shadowed areas.

This system does not require any light-emitting mechanisms on the rotating object itself, and allows users to manipulate the rotating object freely. This enables the system to transform commercially available portable rotating objects, such as hand crank fans or desk fans, into displays without any modification. By utilizing such portable rotating objects, this system can be employed as a spatial interface, facilitating interactions such as pointing a 3D object.

4 CONCLUSION

In this paper, we proposed a method for displaying 3D images using a dynamic projection mapping technique onto a rotating object. Our prototype system demonstrates that layered images can be projected onto user-manipulated rotating objects.

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REFERENCES

- [1] G. E. Favalora, J. Napoli, D. M. Hall, R. K. Dorval, M. G. Giovinco, M. J. Richmond, and W. Chun. 100-million-voxel volumetric display. In *SPIE Defense + Commercial Sensing*, 2002. 1
- [2] T. Fukaya, T. Iwai, and Y. Yamanouchi. Morphovision. In *ACM SIGGRAPH 2006 Emerging Technologies*, SIGGRAPH '06, p. 22–es. Association for Computing Machinery, New York, NY, USA, 2006. doi: 10.1145/1179133.1179156 1
- [3] H. Katsuyama, S. Miyafuji, A. Kohtani, and H. Koike. Volume fan: Turn your layered-fan into a volumetric display. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction*, SUI '23. Association for Computing Machinery, New York, NY, USA, 2023. doi: 10.1145/3607822.3618006 1
- [4] M. Kiyokawa and N. Hashimoto. Dynamic projection mapping with 3d images using volumetric display. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 597–598, 2021. doi: 10.1109/VRW52623.2021.00181 1
- [5] K. WAKATSUKI, C. FUJIKAWA, and M. OMODANI. Universal angle visibility realized by a volumetric 3d display using a rotating mirror-image helix screen. *IEICE Transactions on Electronics*, E107.C(2):23–28, 2024. doi: 10.1587/transele.2023DII0003 1
- [6] T. Yoshida, Y. Watanabe, and M. Ishikawa. Phyxel: Realistic display of shape and appearance using physical objects with high-speed pixelated lighting. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*, UIST '16, p. 453–460. Association for Computing Machinery, New York, NY, USA, 2016. doi: 10.1145/2984511.2984548 1